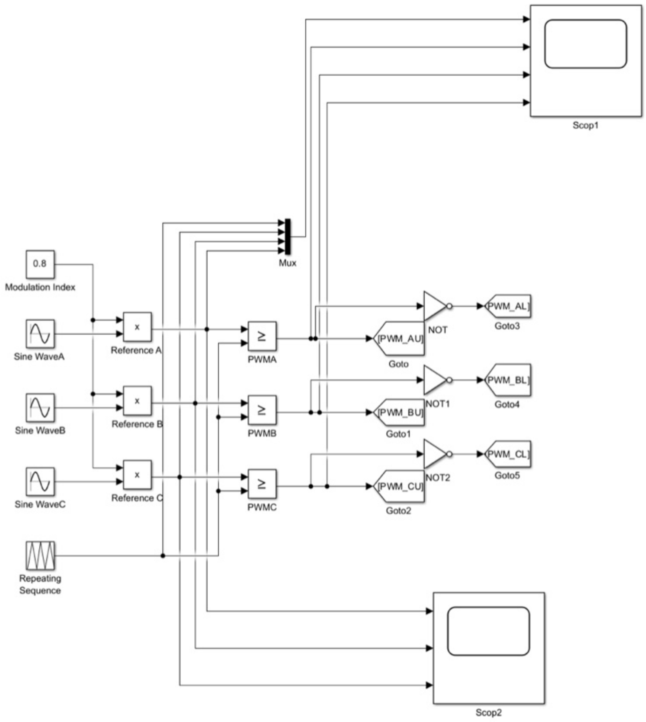
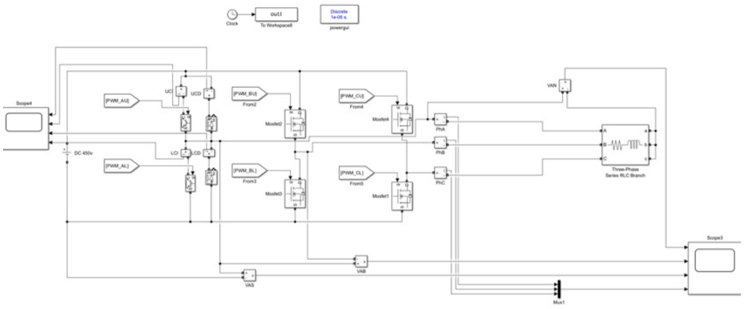
PROJECT 4



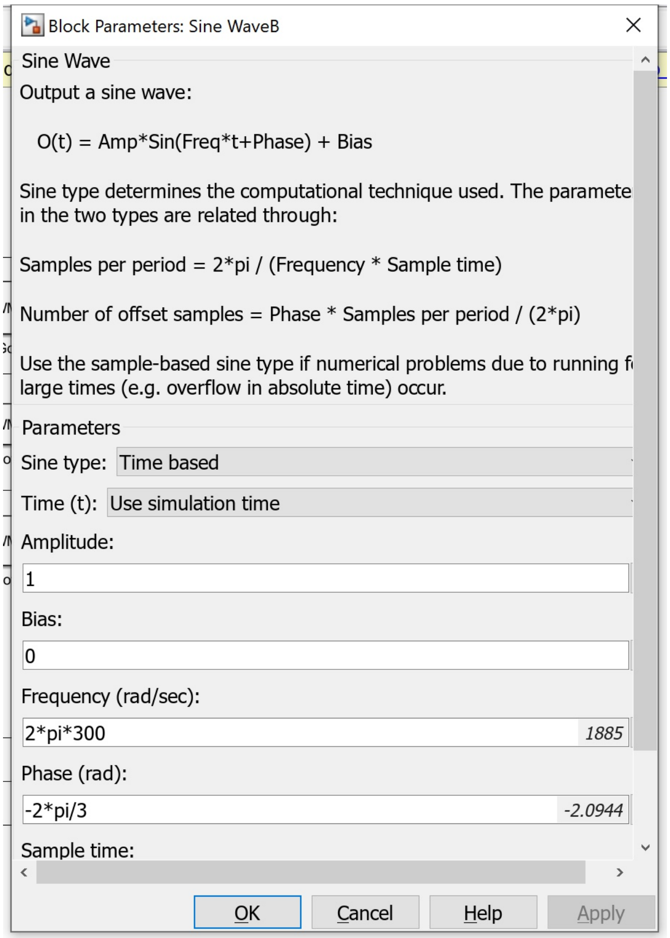
Control / Modulation Stage

This diagram illustrates the control section of the SPWM inverter. For the three phases (A, B, and C) of an inverter, three sinusoidal reference waves will be produced; all three reference waves are 120 degrees apart. The modulation index has been set to 0.8. As such, the amplitudes of the reference waves will scale down by 0.8. The triangle carrier signal will be produced by using a repeating sequence block. To create the PWM signals for the upper switches, the reference signal from each sine wave is compared with the carrier signal using a relational operator. The NOT blocks will then produce the opposite PWM signals to drive the lower switches. The control structure of this method follows the standard sinusoidal PWM strategy for a three-phase inverter.



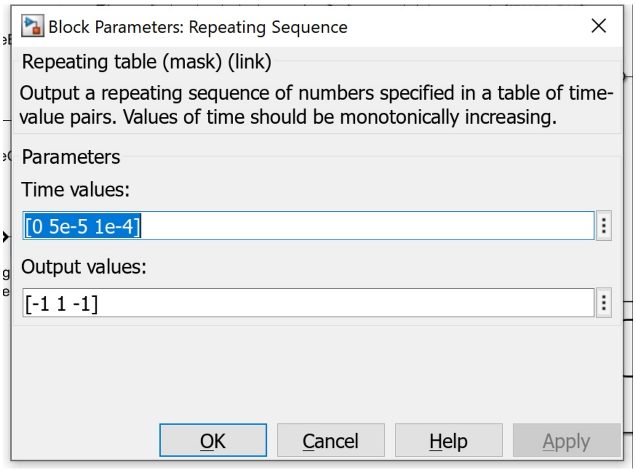
Power Stage

The inverter system's power stage is shown in this figure. The inverter's input is a 450V DC source. The inverter has three legs for each phase A, B, and C. A single IGBT and a diode were added to one phase leg to modify the legs for independent device current measurement. The three-phase RL load at the output represents the equivalent circuit of a motor. Additional blocks were added to the original model to allow observation of pole voltages, line-to-neutral and line-to-line voltages, phase currents, and device currents. The entire configuration is a standard three-phase DC-AC inverter.



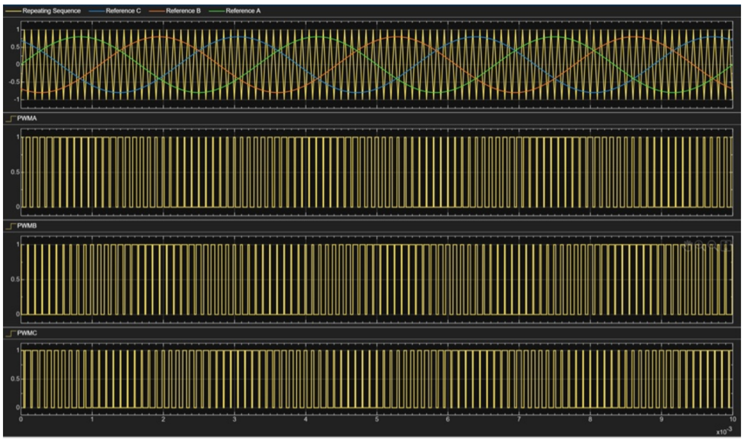
Phase-B Sine Wave Parameter Window

The Phase-B sine wave block is configured as a time-based sine wave. Its amplitude is set to 1 and the bias is 0. The angular frequency is set to 2*pi*300, which corresponds to the required 300 Hz fundamental frequency. The phase is set to -2*pi/3, so Phase B lags Phase A by 120 degrees. This setting is correct for generating a balanced three-phase reference system.



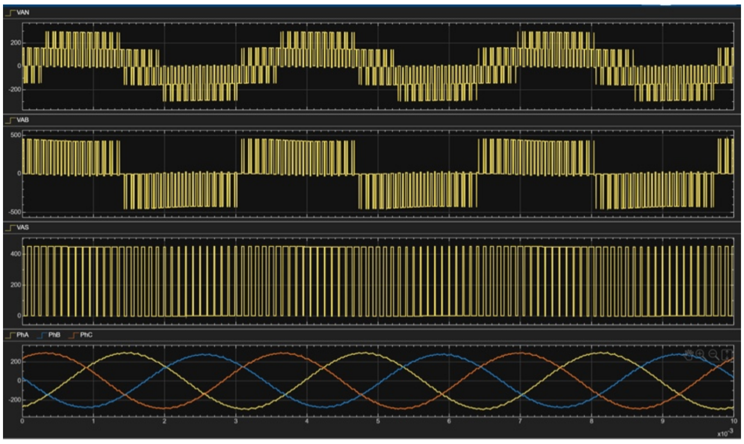
Triangle Carrier Parameter Window

The triangle carrier is generated by the Repeating Sequence block. The time values are set to [0, $5*10^{-5}$,\ $1*10^{-4}$], and the output values are set to [-1,1,-1]. This creates a symmetric triangular waveform with a period of $1*10^{-4}$ s, which corresponds to a switching frequency of 10 kHz. The carrier signal is then used to compare with the three reference sine waves and generate the PWM pulses.



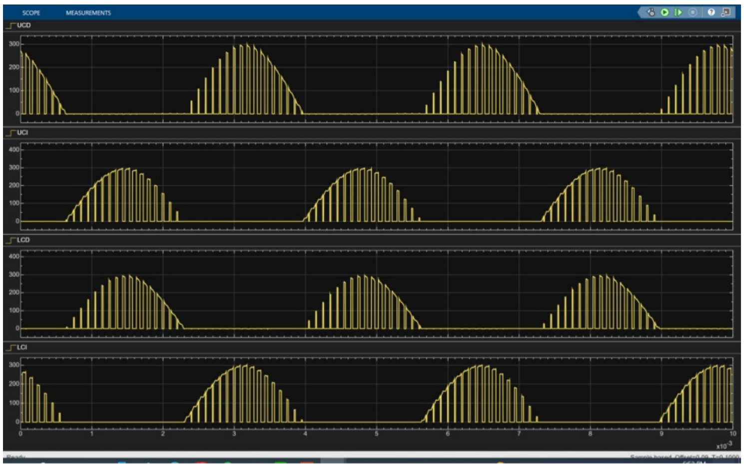
Reference/Carrier Comparison and Three-Phase PWM

The image depicts the triangle wave carrier and 3 sinusoids for the reference signal. It also shows the 3-PWM signals generated for each phase (A, B, and C). Each phase's reference signal is also balanced (i.e., equal in magnitude) but out of phase by 120 degrees. When measuring the pulse width against the carrier, it is found that as the sinusoidal reference signal rises, the pulse width increases and as the sinusoidal reference signal decreases, the pulse width decreases. Therefore, this further demonstrates that the inverter uses sinusoidal PWM operation. Since each phase’s reference sinusoidal signal is different, it stands to reason that each of the generated PWM signals will also be different.



Pole Voltage, Line-Neutral Voltage, Line-Line Voltage, and Three-Phase Currents

This illustration provides the Pole voltage for Phase 'A', Line-neutral voltage for Phase 'A', Phase 'AB' line to line voltage and three phase output currents all plotted on a common time reference. The Pole voltage is a two level switching signal transitioning between the two DC bus levels which is what you expect from an inverter switching node. Line-neutral voltage has both PWM switching characteristics in it but does maintain the fundamental sinusoidal characteristic of the line-neutral voltage. The magnitude of line-line voltage is greater than for either line-neutral voltages; additionally, line-line voltage has step switching characteristic which you would expect from a three phase inverter. The three phase currents are in equilibrium with one another and are sinusoidal with 120 degree phase displacement and indicate the inverter and load are working as intended in steady state conditions. These findings agree with the anticipated performance of a single-phase inverter feeding an appropriately matched balanced three phase RL load while utilizing sinusoidal pulse width modulation (SPWM).



Upper-A and Lower-A Device Currents

This diagram illustrates the various electrical currents that flow through the Upper Diode, Upper Insulated Gate Bipolar Transistor (IGBT), Lower Diode, and Lower IGBT. The electrical current is not constant; rather, it presents an oscillatory nature, due to the on-off electrical operation of the electrical switching devices.

The electrical current envelopes follow the trend of the phase electrical current while each of the currents in the respective individual devices exhibits current flow during various electrically conductive and non-conductive intervals based on the electrical switch's state and freewheeling condition.

The IGBT and diode currents alternate in accordance with the electrical switching logic and electrical current direction. These observations validate the accurate sharing of the current between the IGBT and diode of each upper and lower device.

Overall Discussion

The Simulation Results are similar to one another; however, they possess great importance because they have verified the Control Stage of the PMLC is functioning as expected through both the use of the Reference Signals, Carrier Signals, and Pulse Width Modulated (PWM) Signals. Furthermore, the waveforms that were created from each of the outputs of the three-phase inverter used to measure the waveforms demonstrated the expected switching characteristics of a three-phase inverter. Moreover, the three-phase currents created during the normal operation of the inverter were balanced and had nearly sinusoidal characteristics, indicating that there was sufficient and proper operation of the RL Load. Likewise, the current measurements taken during the operation of the IGBT and Diode Devices exhibited realistic characteristics for conduction consistent with what would be expected from an actual application.

Any discrepancies shown on the Plots created by the computer software can be attributed to incorrect measurement block wiring, Incorrect Polarity of voltage measurement blocks, Incorrect Scope Scaling, or a mismatch of PWM and Carrier Settings; additionally, the use of the incorrect reference node during the measurement of Longitudinal-neutral Voltage may have contributed. Overall, the Final Model demonstrates expected performance characteristics of the Three- Phase SPWM Inverter.